

Blanka Balogh

Research Scientist | Toulouse, France

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SUMMARY

Research scientist specializing in hybrid physics/ML systems for geophysical and environmental modeling, with applications to weather, climate, ocean, and wildfire prediction. Experienced in designing and deploying scientific ML methods at scale on HPC infrastructure, and in leading research end-to-end, from hypothesis formulation to operational testing and publication.

TECHNICAL SKILLS AND METHODS

Deep Learning Frameworks: PyTorch, Lightning, Tensorflow, Keras, MLFlow, Tensorboard, anemoi

Scientific Computing and HPC: MPI, openMP, Slurm, Linux, Shell, Fortran, Python, (py)OASIS, git

Research Methods: neural network parameterizations, hybrid physics/ML, probabilistic modeling, scientific ML, scientific computing

EXPERIENCE

Research Scientist

Centre National de Recherches Météorologiques, Météo-France

Oct 2022 – present

Toulouse, France

- Led research projects on AI for multiscale geophysical systems and hybrid climate modeling using the ARP-GEM global climate model
- Developed hybrid physics/ML methods for weather and climate modeling using OASIS and PhyDLL
- Collaborated with international research groups on probabilistic modeling and benchmarking hybrid physics/ML climate simulations
- Built ML workflows for atmospheric and ocean modeling using PyTorch, anemoi and HPC infrastructure in ARP-GEM
- Worked on simulation acceleration and optimization for hybrid climate workflows leveraging heterogeneous HPC partitions
- Mentored interns and advised ML components of interdisciplinary research projects

PI of project GAIAC – French National Program INSU-SUN

Subject: ML parameterizations with improved out-of-sample generalizability

Jan 2026 – Dec 2028

Visiting scholar

Pierre Gentine's group, Columbia University

Sept 2023 – Dec 2023

New York, NY, USA

EDUCATION AND TRAININGS

1-week in-person ECMWF training courses

'Parameterizations of subgrid processes' (2025) and 'Numerical methods for weather prediction' (2024) Reading, UK

Institut National Polytechnique de Toulouse

PhD in Earth and Planetary sciences. Subject: Hybrid climate modeling

Sept 2019 – Oct 2022

Toulouse, France

Ecole Nationale de la Météorologie

M.Sc.Eng. in Weather Forecasting and Numerical Modeling

Sept 2016 – Aug 2019

Toulouse, France

INSA Toulouse

M.Sc. exchange year in Applied Mathematics with a major in machine learning

Sept 2018 – Aug 2019

Toulouse, France

CPGE PCSI/PC* Lycée Henri IV, Paris, France

Sept 2014 – Aug 2016

SELECTED PUBLICATIONS

- Garcia-Cristobal, J., Wurtz, J., Doury, A., **Balogh, B.**, Masson, V., and Mestre, O., 2026: A Neural Network-based downscaling method for near surface temperature in urban areas. Under review at Urban Climate.
- Germain, H., **Balogh, B.**, Geoffroy, O., Saint-Martin, D., 2025: Improvement of a neural network convection scheme by including triggering and evaluation in present and future climates. Under review at Artif. Intell. Earth Syst.
- **Balogh, B.**, D. Saint-Martin, and O. Geoffroy, 2025: Online Test of a Neural Network Deep Convection Parameterization in ARP-GEM1. Artif. Intell. Earth Syst.
- **Balogh, B.**, Saint-Martin, D., and Ribes, A., 2022: How to calibrate a dynamical system with neural network based physics? Geophysical Research Letters.
- **Balogh, B.**, Saint-Martin, D., and Ribes, A., 2021: A toy model to investigate stability of AI-based dynamical systems. Geophysical Research Letters.

SELECTED PRESENTATIONS

- Invited Speaker: CNRS annual conference 'Défis Théoriques pour la science du climat', Paris, France (2026).
- Oral presentation: EGU26, Vienna, Austria (2026).
- Poster presentation: Exclaim! Symposium, Zürich, Switzerland (2025).
- Oral presentation: EGU24, Vienna, Austria (2024).

SCIENTIFIC SERVICE AND LEADERSHIP

Co-organizer of the annual AI Days at Météo-France ('Journées de l'IA', 3 editions) <i>Internal conference gathering 200+ participants from across the organization and external speakers.</i>	2024 – present
Associate Editor <i>Artificial Intelligence for the Earth Systems (American Meteorological Society)</i>	2023 – present
Co-convener at EGU26 <i>Session: Machine Learning for climate science, 50+ oral presentations and posters.</i>	2026
External Reviewer <i>External Reviewer for a French National Research Agency (ANR) project</i>	2026
Scientific Program committee member for ICML <i>Reviewer for the workshop 'Machine Learning for Earth System Modeling'</i>	2024

RESEARCH PARTNERSHIPS

- Probabilistic multi-fidelity parameterization for subgrid-scale climate processes in collaboration with Columbia University (P. Gentine's group).
- Machine learning approaches for global wildfire burned areas prediction with CNRM/CNRS collaborators.
- Contributed to the development and debugging of open-source coupling tools for hybrid physics/ML climate workflows (PhyDLL, CERFACS collaboration)

RESEARCH SUPERVISION

- ML advisor for PhD projects on climate downscaling, urban climate modeling and wildfire prediction
- Supervised M.Sc. and research internship projects on stochastic and physically consistent climate modeling